

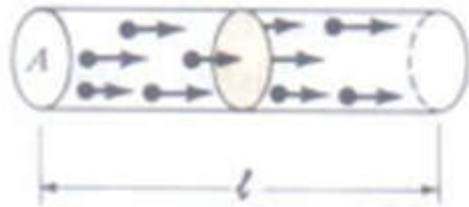
Chapter 17 Direct Currents

In this chapter we will study:

- **Electric current and current density**
- **Electric Resistance**
- **Kirchhoff's Rules: single and multi-loop circuits**
- **Energy sources: electromotive force emf, internal resistance and terminal voltage**
- **Power**
- **Series and parallel resistors**
- **Galvanometers, Voltmeters and Ammeters**
- **RC-Circuits: Charging and discharging a capacitor**

DIRECT CURRENTS

The electric current in a wire is rate at which charges moves in the wire.



ΔQ =charge passes
through cross-sectional
area A
 Δt =time required to pass A
(See Fig.)

The average current is

$$\bar{I} = \frac{\Delta Q}{\Delta t}$$

The instantaneous current is

$$I = \frac{dQ}{dt}$$

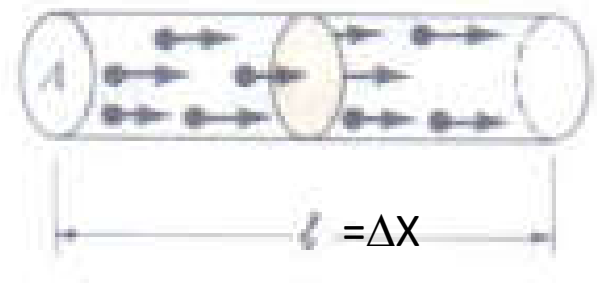
Example 7.1: An electrochemical cell consists of two electrodes placed in an aqueous solution of silver nitrate. A constant 0.5A current is passed through the cell for 1 hour. Find the total charge transported through the cell in coulombs and multiples of electronic charge.

Solution:

$$\begin{aligned}\Delta Q &= I\Delta t = 0.5A \times 1h \\ &= 0.5A \times 60 \times 60S \\ &= 1800C\end{aligned}$$

Current density **$J = I/A$** ,
 A is the cross-sectional area
 So,

$$I = \frac{\Delta Q}{\Delta t} = \frac{enlA}{l/V_d} = enAV_d$$



V_d is the drift speed, **e** is electron charge
 n carriers number (charge per unit volume).

$$I = \Delta Q / \Delta t = Nq A \Delta x / \Delta t \quad A \Delta x = V \\ = (N/V)qAV_d = nqAV_d$$

V_d = drift speed
 A = cross-sectional area
 N = # of charge carries
 V = Volume
 $q = e$

Example 7.2: Number 12 copper wire is often used to wire household electrical outlets. It is radius 1mm. If it carries 10A, what is the drift speed of the electrons (atomic mass of Cu=64u and its density 8900kg/m³)

To find n, the density d and mass M

NOTE: 1u = 1.66×10⁻²⁷kg, u = atomic mass unit



$$n = d/M$$

$$n = \frac{d}{M} = \frac{8900 \text{ kg m}^{-3}}{(64 \text{ u})(1.66 \times 10^{-27} \text{ kg u}^{-1})}$$

$$= 8.38 \times 10^{28} \text{ m}^{-3}$$

$$v = \frac{I}{neA} = \frac{I}{ne\pi r^2}$$

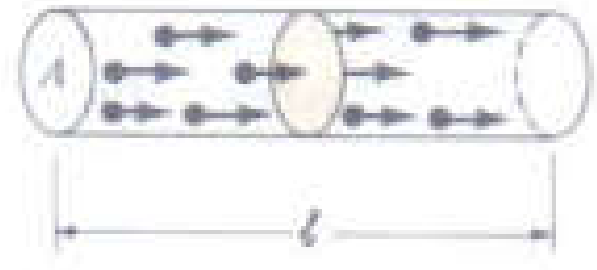
$$= \frac{10 \text{ A}}{(8.38 \times 10^{28} \text{ m}^{-3})(1.6 \times 10^{-19} \text{ C})\pi(10^{-3} \text{ m})^2}$$

$$= 2.37 \times 10^{-4} \text{ m s}^{-1}$$

17.2 | RESISTANCE

$$R = \frac{V}{I}$$

V= potential difference
I=electric current



$$R = \frac{\rho l}{A}$$

$$J = \sigma E, \xrightarrow{\text{yields}} \frac{I}{A} = \frac{1}{\rho} \frac{V}{L} \rightarrow V = \left(\frac{\rho l}{A} \right) I$$

ρ is the resistivity,

L=length, A=corss-sectional area

σ is the conductivity (table 17.1)

$$\sigma = \frac{1}{\rho}$$

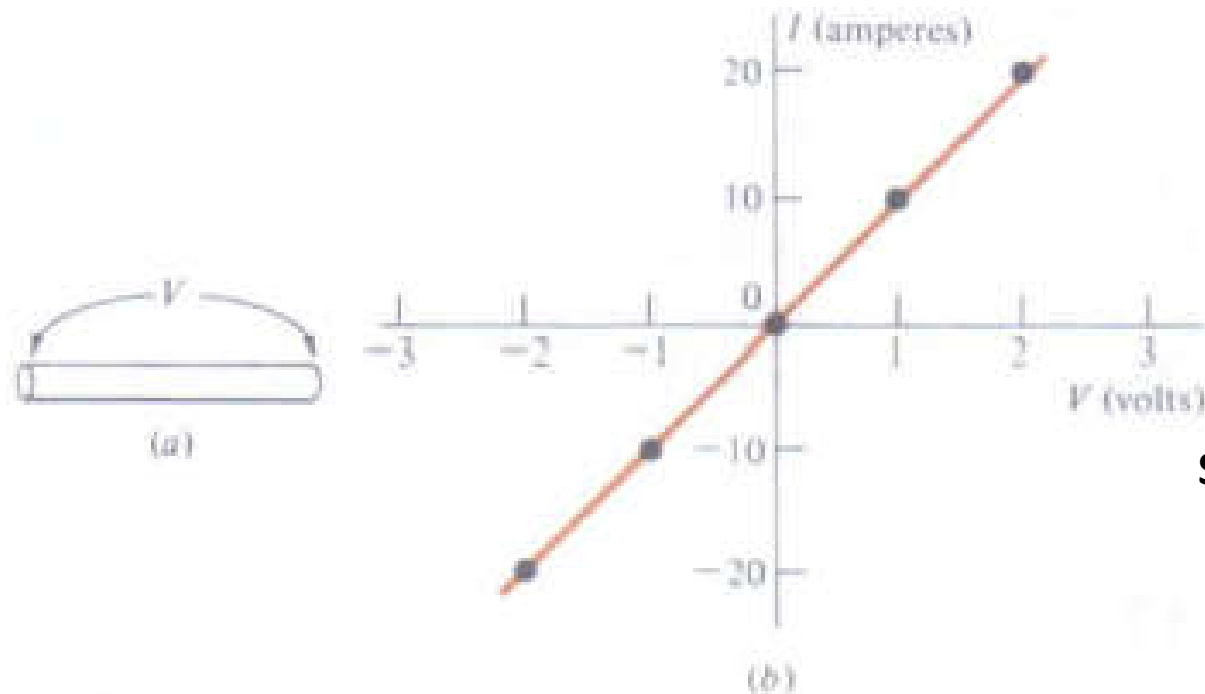
V = IR **Ohm's Law**

TABLE 17.1**Resistivities at 20°C in ohm-metres**

	Substance	Resistivity (ohm-metres)
Conductors	Silver	1.47×10^{-8}
	Copper	1.72×10^{-8}
	Aluminum	2.63×10^{-8}
Semiconductors	Germanium	0.60
	Silicon	2300
Insulators	Sulfur	10^{15}
	Glass	10^{10} – 10^{14}
Ionic conductors	Body fluids	approx. 0.15

Example 17.3

Find the resistance of the wire in Fig. 17.2.



Solution: Slope equal I/V

$$R = \frac{V}{I} = \frac{1 \text{ V}}{10 \text{ A}} = 0.1 \text{ ohm}$$

Figure 17.2. (a) A copper wire has a variable potential difference V between its ends. (b) The current varies linearly with the potential difference in the wire, so it is an ohmic conductor.

Example 17.4

Find the room temperature resistance of a copper wire 100 m long with a radius of 1 mm = 10^{-3} m.

According to Table 17.1, the resistivity of copper at room temperature (20° C) is 1.72×10^{-8} ohm m. Hence the resistance of this wire is

$$\begin{aligned} R &= \frac{\rho \ell}{A} = \frac{\rho \ell}{\pi r^2} \\ &= \frac{(1.72 \times 10^{-8} \text{ ohm m})(100 \text{ m})}{\pi (10^{-3} \text{ m})^2} \\ &= 0.547 \text{ ohm} \end{aligned}$$

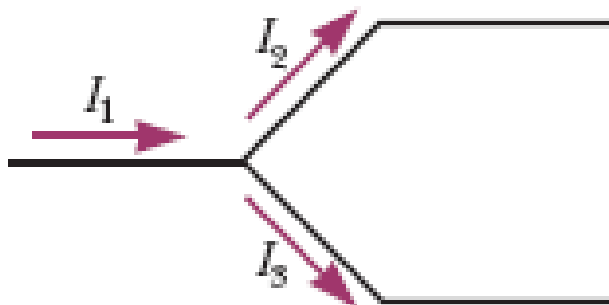
Kirchhoff's Rules

1. **Junction rule.** At any junction, the sum of the currents must equal zero:

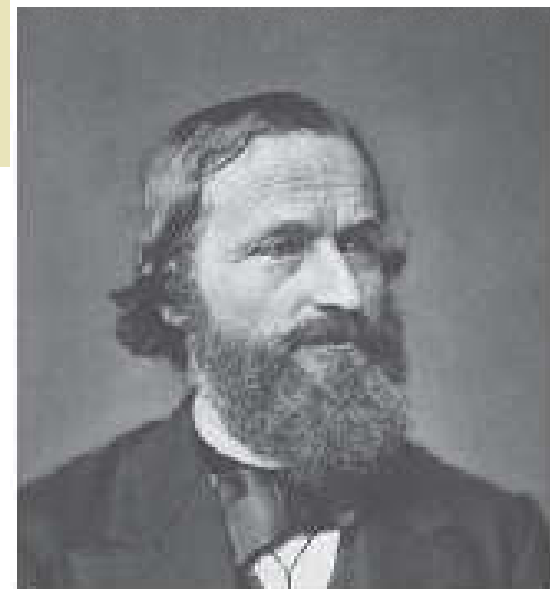
$$\sum_{\text{junction}} I = 0$$

2. **Loop rule.** The sum of the potential differences across all elements around any closed circuit loop must be zero:

$$\sum_{\text{closed loop}} \Delta V = 0$$



$$I_1 - I_2 - I_3 = 0$$



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Gustav Kirchhoff

German Physicist (1824–1887)

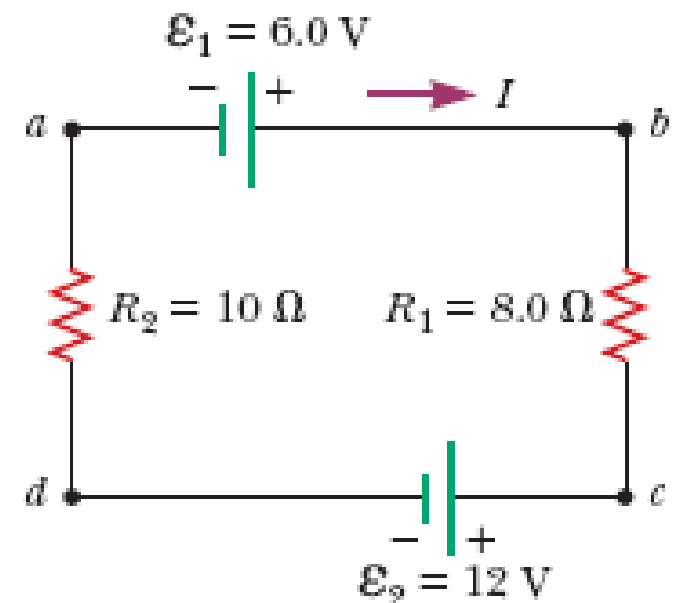
A Single-Loop Circuit

A single-loop circuit contains two resistors and two batteries as shown in Figure . (Neglect the internal resistances of the batteries.) Find the current in the circuit.

SOLUTION

$$\sum \Delta V = 0 \rightarrow \mathcal{E}_1 - IR_1 - \mathcal{E}_2 - IR_2 = 0$$

$$I = \frac{\mathcal{E}_1 - \mathcal{E}_2}{R_1 + R_2} = \frac{6.0 \text{ V} - 12 \text{ V}}{8.0 \, \Omega + 10 \, \Omega} = -0.33 \text{ A}$$

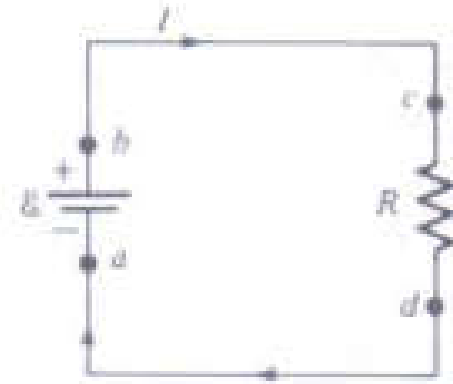


A series circuit containing two batteries and two resistors, where the polarities of the batteries are in opposition.

17.3 | ENERGY SOURCES IN CIRCUITS

$$\varepsilon = IR$$

ε is the electromotive force



Example 17.5

A dry cell with an EMF of 6 V is connected to a light bulb with resistance 4 ohms. Find the current.

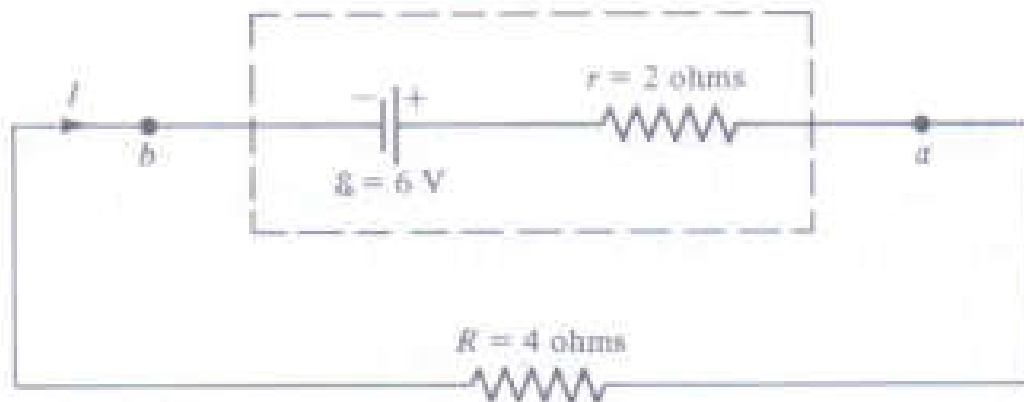
With $\varepsilon = IR$, the current is

$$I = \frac{\varepsilon}{R} = \frac{6 \text{ V}}{4 \text{ ohms}} = 1.5 \text{ A}$$

Internal resistance r and terminal voltage

Example 17.6

A battery with an EMF of 6 V and an internal resistance $r = 2$ ohms is connected to an $R = 4$ -ohm light-bulb (Fig. 17.7). Find (a) the current; (b) the terminal potential difference of the battery.



$$\mathcal{E} = IR + Ir \quad \longrightarrow \quad I = \frac{\mathcal{E}}{R + r} = \frac{6 \text{ V}}{(2 + 4) \text{ ohms}} = 1 \text{ A}$$

(b) The terminal voltage is the potential difference V_{ab} between points a and b .

$$V_{ab} = \mathcal{E} - Ir = 6 \text{ V} - (1 \text{ A})(2 \text{ ohms}) = 4 \text{ V}$$

□NOTE: $\mathcal{E}$ is the emf of the battery

V_{ab} is the terminal voltage across the battery

17.4 | POWER IN ELECTRICAL CIRCUITS

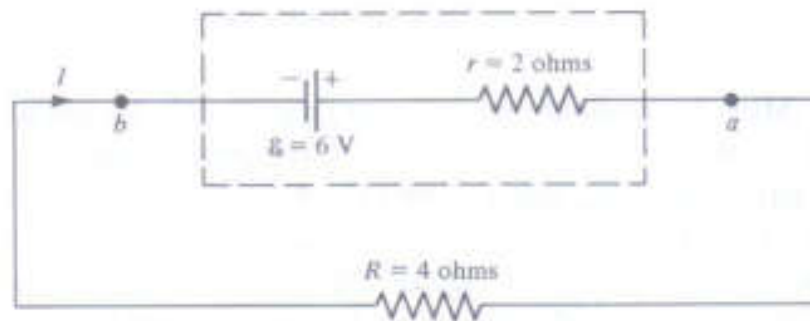
The power is the rate of work

$$\mathcal{P} = \frac{\Delta W}{\Delta t} = IV$$

P is the power

$$V = IR, \text{ and } \mathcal{P} = I^2 R$$

Example: Calculate the power supplied to or by each element in the circuit shown



$I = 1$ A from previous example

$$P_{(2\Omega)} = 1^2 \times 2 = 2 \text{ W}$$

$$P_{(4\Omega)} = 1^2 \times 4 = 4 \text{ W}$$

$$P(\varepsilon) = I \times \varepsilon = 1 \times 6 = 6 \text{ W}$$

□ NOTE: Power supplied = power dissipated

$$\begin{aligned} 1 \text{ kW h} &= (10^3 \text{ W})(3600 \text{ s}) \\ &= 3.6 \times 10^6 \text{ J} \end{aligned}$$

Example 17.9

For a household 60-W light bulb operated at 120 V, find (a) the current; (b) the resistance; (c) the 24-hour operating cost, if energy costs 10 cents per kilowatt hour.

(a) Since the power $\mathcal{P}$ in any circuit element equals VI ,

$$I = \frac{\mathcal{P}}{V} = \frac{60 \text{ W}}{120 \text{ V}} = 0.5 \text{ A}$$

(b) The resistance is

$$R = \frac{V}{I} = \frac{120 \text{ V}}{0.5 \text{ A}} = 240 \text{ ohms}$$

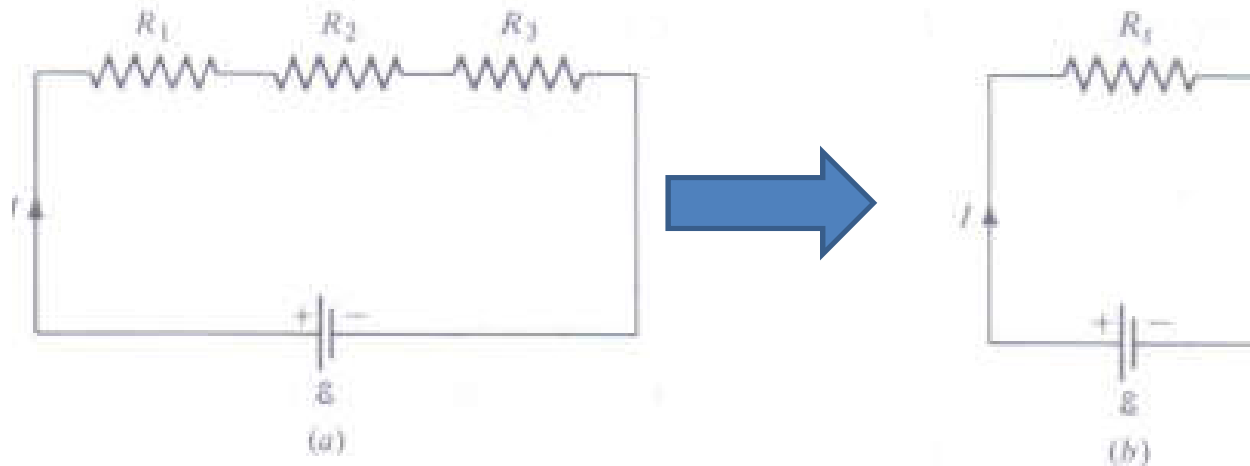
(c) The bulb uses power at the rate of 60 W or 0.060 kW. Thus the cost of operating it for 24 hours is

$$(0.06 \text{ kW})(24 \text{ h})[10\text{¢} (\text{kW h})^{-1}] = 14.4\text{¢}$$

17.5 | SERIES AND PARALLEL RESISTORS; KIRCHHOFF'S RULES

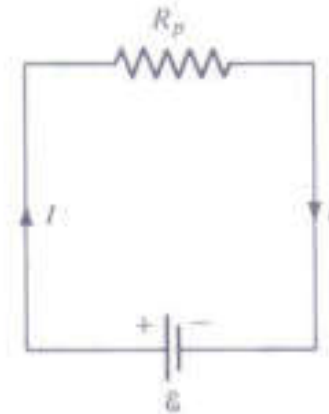
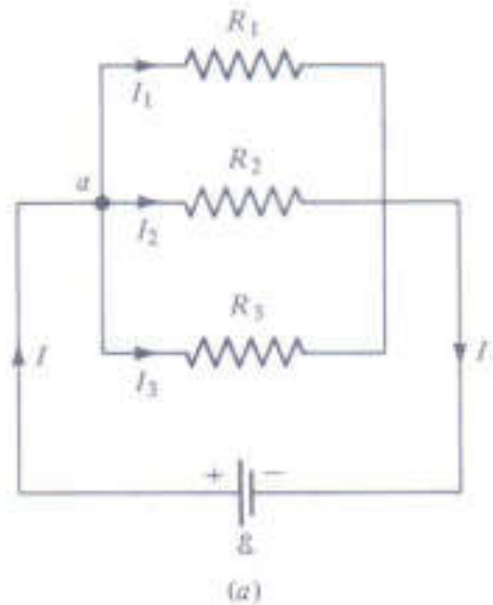
Series: $R_s = R_1 + R_2 + R_3 + \dots$

Same current I



$$\varepsilon - IR_1 - IR_2 - IR_2 = 0 \Rightarrow I = \frac{\varepsilon}{R_1 + R_2 + R_3}$$

Parallel: $1/R_p = 1/R_1 + 1/R_2 + 1/R_3 + \dots$
Same potential



$$I = I_1 + I_2 + I_3 \quad \Rightarrow \quad I = \frac{\varepsilon}{R_1} + \frac{\varepsilon}{R_2} + \frac{\varepsilon}{R_3}$$
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Example 17.10

(a) Find the equivalent resistance of the resistors in Fig. 17.12a. (b) Find the current in each resistor.

(a) The current I splits at point a into two currents, I_1 and I_2 . I_1 goes through each of the 1-ohm resistors, so they are in series. Their equivalent resistance is then

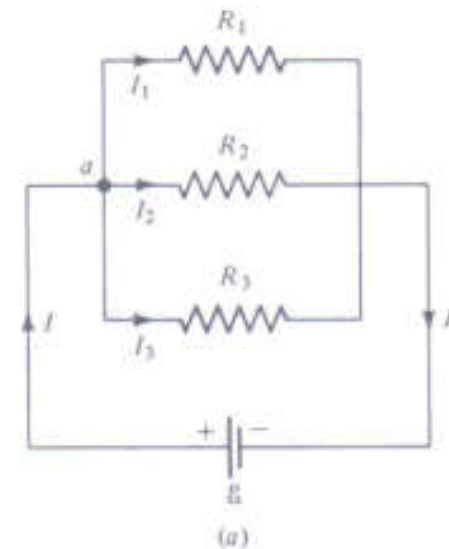
$$R_s = 1 \text{ ohm} + 1 \text{ ohm} + 1 \text{ ohm} = 3 \text{ ohms}$$

This reduces the circuit to two parallel 3-ohm resistors (Fig. 17.12b). The effective resistance of the entire system is found from

$$\frac{1}{R_p} = \frac{1}{3 \text{ ohms}} + \frac{1}{3 \text{ ohms}} = \frac{2}{3 \text{ ohms}}$$

Thus

$$R_p = 1.5 \text{ ohms}$$



17.6 | VOLTMETERS AND AMMETERS

Galvanometer

A galvanometer is a device that is used to detect small electric current or measure its magnitude

Galvanometer has small resistance R_G
Small current I_G



VOLTMETER

A voltmeter is an instrument used for measuring electric potential difference between two points in an electric circuit

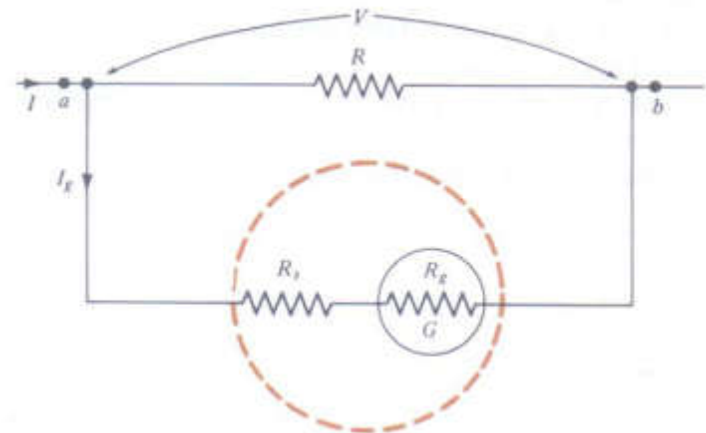


Voltmeter is a Galvanometer connected in series with large resistance R_s

So, $V = I_g(R_g + R_s)$

To measure a potential difference between two points in a electric circuit
The voltmeter is connected in parallel

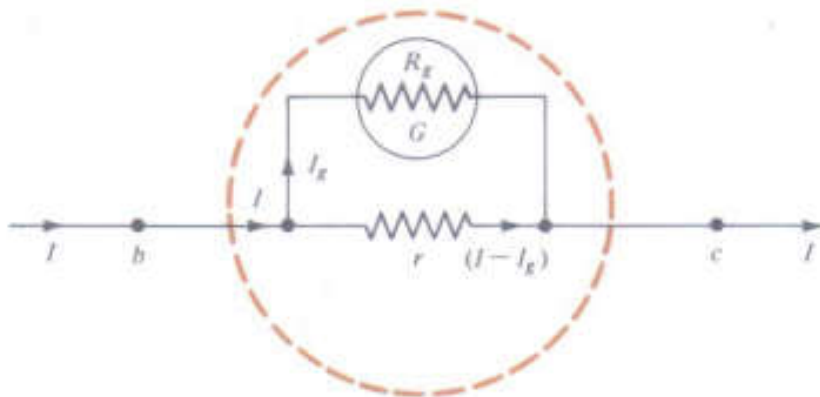
$$V = I_g(R_g + R_s)$$



Ammeter

An ammeter (abbreviation of Ampere meter) is an instrument used to measure the current in a circuit

The ammeter is a Galvanometer connected in with a small resistance R_p in parallel



$$V = I_g R_g = (I - I_g) r$$

Example 17.11: A galvanometer with a resistance of 100Ω gives a full-scale deflection for a current of 1mA .

(a) How large a parallel resistor is needed to convert it into ammeter with a 20A range?

(b) What is the ammeter resistance.

Solution:

$$(a) \ r = \frac{R_g I_g}{I - I_g} = \frac{100\Omega \times 0.001\text{A}}{20\text{A} - 0.001\text{A}} = 0.005\Omega$$

$$(b) \ r \ll R_g$$

$$\frac{1}{R_A} = \frac{1}{r} + \frac{1}{R_g} = \frac{1}{0.005} + \frac{1}{100}$$

$$R_A = 0.005\Omega$$

Example 17.12: A galvanometer of example 17.11 has a resistance of 100Ω and gives a full-scale deflection for a current of 1mA .

(a) How can be converted into a voltmeter with 100 V range?

(B) What is the voltmeter resistance?

Solution:

(a) The series resistance needed is

$$R_s = \frac{V}{I_g} - R_g = \frac{100}{0.001} - 100 = 99900\Omega$$

$$\begin{aligned}(b) R_V &= R_s + R_g \\ &= 99900 + 100 \\ &= 100000\Omega\end{aligned}$$

17.7 | CIRCUITS CONTAINING RESISTANCE AND CAPACITANCE

RC-Circuits

Charging a Capacitor:

$$\varepsilon - IR - \frac{q}{C} = 0$$

$$I = \frac{\varepsilon}{R} - \frac{q}{RC} = \frac{dq}{dt}$$

$$\Rightarrow q = q_f(1 - e^{-t/RC})$$

$$V = V_f(1 - e^{-t/RC})$$

$$I = I_o e^{-t/RC}$$

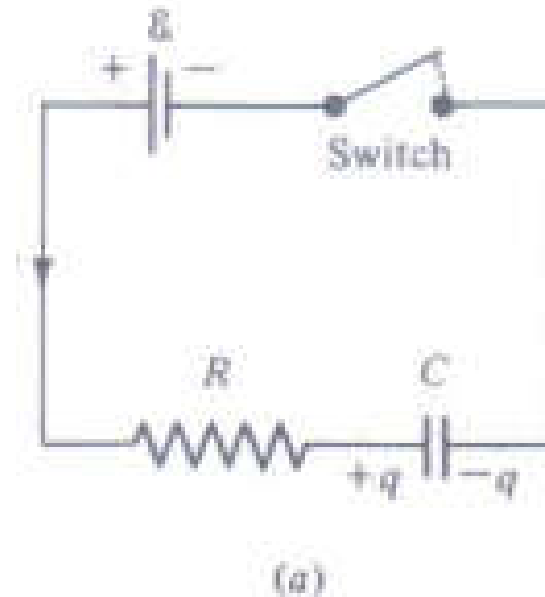
q is the charge on the capacitor at any time

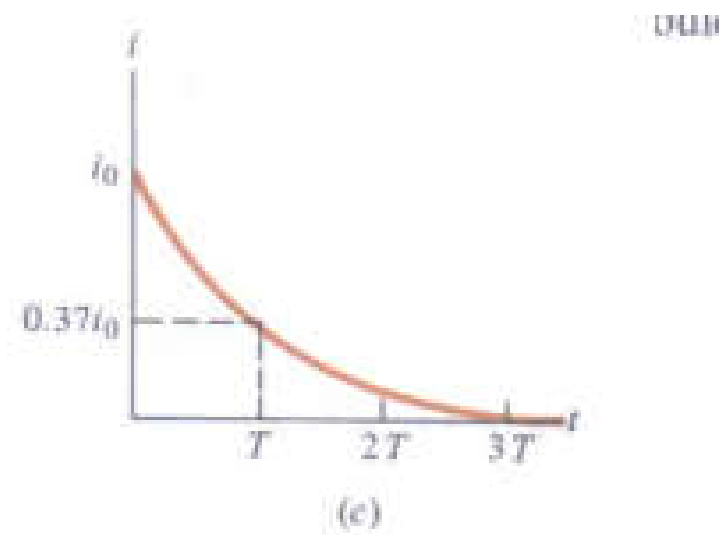
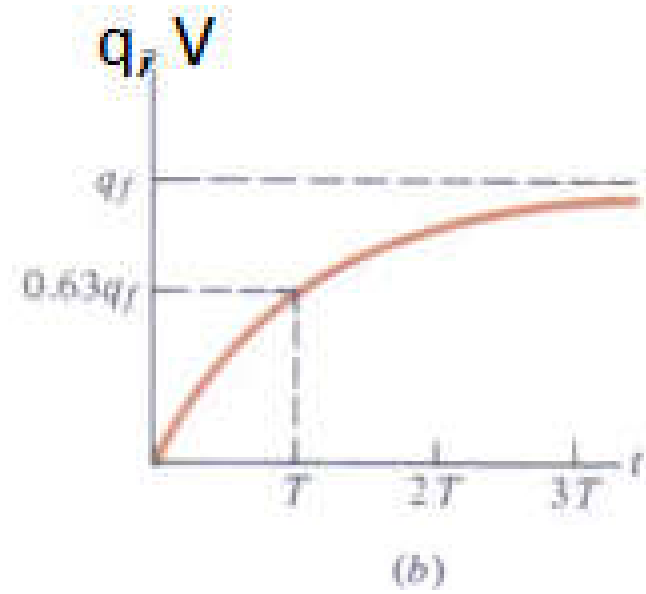
I is the current at any time, R is load resistance

t is the time

ε is the emf of the battery, q_f is final charge on the capacitor

$$q_f = \varepsilon C \quad (t = \infty)$$





Example 17.13: In Figure shown $C=2\mu\text{F}$, $R=1000\Omega$, $\varepsilon =6\text{V}$. Find
(a) the final charge on the capacitor. (b) the initial current
(b) (c) the time constant.

Solution:

$$(a) q_f = \varepsilon C$$

$$= 6 \times (2 \times 10^{-6})$$

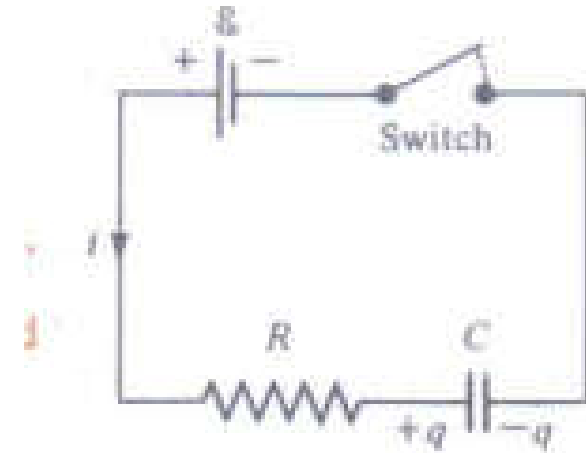
$$= 1.2 \times 10^{-5} \text{C}$$

$$(b) I_o = \frac{\varepsilon}{R} = \frac{6}{1000} = 6\text{mA}$$

$$(c) T = RC$$

$$= 1000 \times (2 \times 10^{-6})$$

$$= 2\text{mS}$$

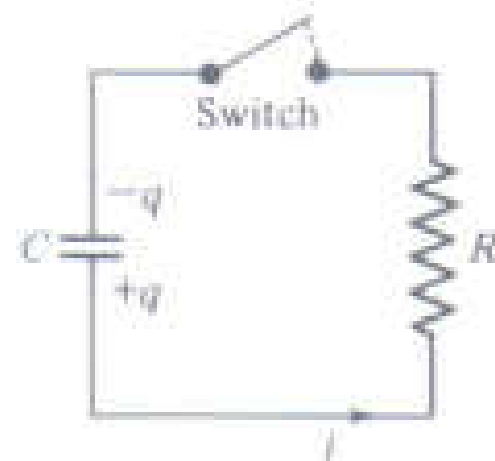
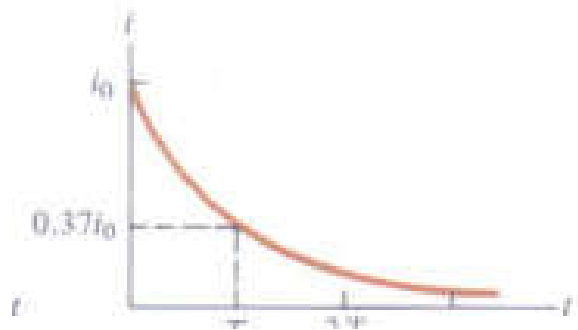
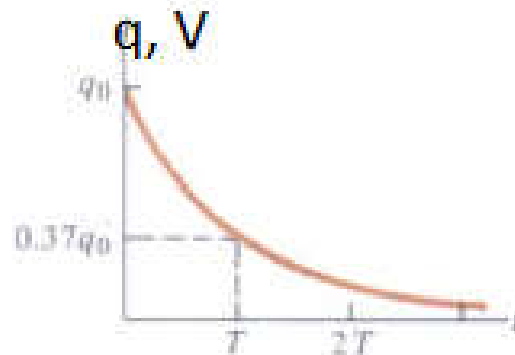


Discharging a Capacitor:

$$V_C = -V_R \Rightarrow \frac{q}{C} = -RI = -R \frac{dq}{dt}$$

$$q = q_0 e^{-t/RC}$$

$$I = I_0 e^{-t/RC}$$



(a)

$T = RC =$ time constant
 q is the charge at any time
 q_0 is the initial charge
 I is the current at any time
 I_0 is the initial current
 R is the load resistance

Example:

A 10^{-4}F capacitor is **discharged** through a 100Ω resistor. How long does it take for the charge on the capacitor to drop to $1/e^2$ times its original value?

Solution:

$$q = q_o e^{-t/RC} = q_o e^{-2}$$



$$t/RC = 2$$



$$t = 2 \times RC = 2 \times 100 \times 10^{-4} = 0.02 \text{ S}$$